

Experience

Apple ▪ Senior Software Engineer (Contract)

System Intelligence Team, Machine Learning and AI Group March 2024 – Sep 2025

- Productionized research prototype [Talaria](#) supporting Apple foundation model latency optimization using Flask, Vue, Svelte, and K8s, improving on-device inference efficiency of 50+ machine learning vision models
- Led design and development of internal Next.js storefronts for configuring, integrating, and sharing LLMs
- Expanded interactive graph visualization tool Mycelium ([open-source](#)), a TypeScript and Svelte library used for visualizing deep neural networks and Transformer architectures. Implemented WebGL viewer achieving 5x higher framerates when viewing LLMs, and enhanced feature set for models running on Apple Neural Engine.
- Engineered Python library enabling researchers to compile models to hardware with high reliability ($\geq 99.9\%$)

Memorial Sloan Kettering Cancer Center ▪ Senior Software Engineer

Pathology Results Team, Clinical Bioinformatics Group May 2018 – Present

- Tech lead for MPath, a clinical pathology software for displaying genomic testing results and matching patients to targeted therapies. Leading 8 developers while establishing roadmaps and collaborations across 5 teams.
- Architected and implemented performant web apps for managing and visualizing high-throughput clinical data supporting 120k+ patients, using React, Flask, MySQL, and MongoDB. Code steward for all front-end releases.
- Redesign and rewrote core UIs, improving query times, performance metrics, code splitting, end-to-end testing, state machines, and application monitoring, achieving 24% faster clinical report turnaround times
- Spearheaded development lifecycle by creating build automations, design system, and CI/CD pipelines for 40+ developers to achieve code optimization, visual unity, and standardized tooling across 10+ portals
- Developed large ETL systems and APIs delivering 2x faster processing times using Python, Airflow, and K8s

Skills

Languages ▪ JavaScript, TypeScript, Python, Ruby, SQL, HTML, CSS, Sass

Libraries ▪ React, Flask, FastAPI, Next.js, Vue, Tailwind, Node.js, Playwright, Cypress, pandas

Tools ▪ Git, Docker, GitLab CI/CD, GitHub Actions, Apache Airflow, nginx, AWS S3, ELK, K8s

Design ▪ Figma, Adobe Photoshop, Lightroom, Illustrator

Education

Macaulay Honors College New York, NY

B.A. Chemistry and Bioinformatics, Minor in Mathematics, GPA 3.95 Aug 2013 – May 2017

- Data Science Fellowship 2016
- Full Merit Scholarship 2013 – 2017

Publications

Khwaja E, Song YS, **Agarunov A**, Huang B. CELL-E 2: Translating Proteins to Pictures and Back with a Bidirectional Text-to-Image Transformer. **NeurIPS** (2023).

Brannon AR, [et al, including **Agarunov A**]. Enhanced specificity of clinical high-sensitivity tumor mutation profiling in cfDNA via paired normal sequencing using MSK-ACCESS. **Nature Communications** (2021).

Projects

Turbokit ▪ Framework for building production-ready, scalable, monorepo microfrontend React apps

ronivonu ▪ Generative art and creative coding projects created in JavaScript, Processing, and Python